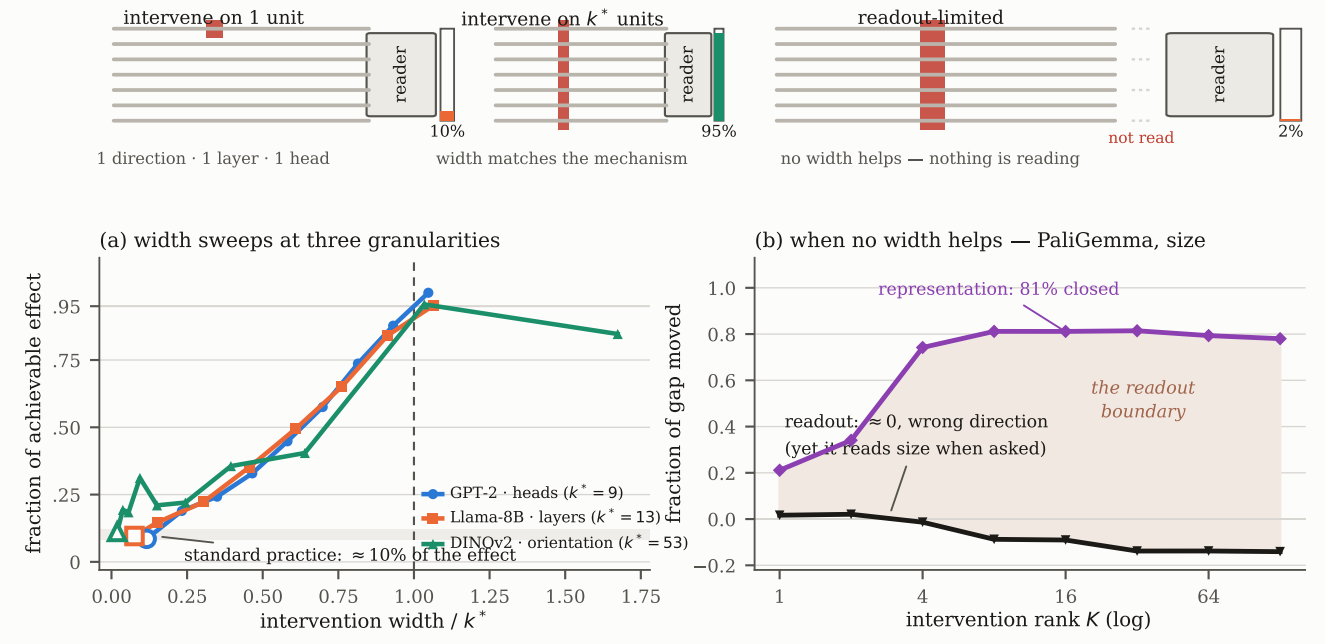
intervene on 1 unit
reader
10%
1 direction · 1 layer · 1 head
intervene on k^* units
reader
95%
width matches the mechanism
readout-limited
reader
not read
2%
no width helps — nothing is reading
(a) width sweeps at three granularities
fraction of achievable effect
.95
.75
.50
.25
0
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
intervention width / k^*
GPT-2 · heads ($k^* = 9$)
Llama-8B · layers ($k^* = 13$)
DINOv2 · orientation ($k^* = 53$)
standard practice: ≈ 10% of the effect
(b) when no width helps — PaliGemma, size
fraction of gap moved
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
representation: 81% closed
the readout boundary
readout: ≈ 0, wrong direction
(yet it reads size when asked)
1
4
16
64
intervention rank K (log)